Rank-Constrained Adaptation Destroys Collaborative Behavior in Multi-Agent LLMs

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Abstract

Multi-agent LLM deliberation has been reported to both improve and harm task accuracy, with recent controlled studies (Du et al., 2023; *Talk Isn't Always Cheap*, 2025; *Can LLM Agents Really Debate?*, 2025) yielding outcomes that span roughly -10 to +15 percentage points on matched benchmarks. We argue that a single under-controlled variable explains a large portion of this spread: the *training method* used to produce the domain specialists participating in the deliberation. We run a controlled comparison in which LoRA-finetuned and fully-finetuned Qwen3 specialists reach identical solo accuracy on their target domain, then engage in a two-agent natural-language collaboration protocol with a matched partner. At 4B parameters and 84% solo accuracy on medicine, full fine-tuning yields a +5.0 pp collaboration delta with a 1.4× correct-to-wrong / wrong-to-correct (C2W/W2C) switch ratio, while LoRA at rank 128 yields a +1.5 pp delta with a 19× C2W/W2C ratio. The LoRA specialist almost never recovers an incorrect peer answer, and it routinely abandons its own correct one. A rank sweep from r=4 to r=128 at both 1.7B and 4B fails to close the gap. We argue the mechanism is LoRA's low-rank update geometry; the "intruder dimensions" it introduces (Shuttleworth et al., 2410.21228) are precisely the directions that dominate collaborative updating. Cheap adaptation has a hidden cost, a collapse of the information channel that multi-agent deliberation relies on.

Keywords: multi-agent LLMs, LoRA, full fine-tuning, deliberation, rank constraint, calibration.

1. Introduction

1.1. Motivation

Multi-agent LLM systems, in which two or more language models exchange natural-language reasoning over several rounds before producing a final answer, are one of the most widely deployed inference-time scaffolds for small and mid-size open-source models. Debate, deliberation, mixture-of-agents, chain-of-experts, and mediator architectures have all been proposed as ways to extract more from a fixed model class without retraining (Du et al., 2023; Liang et al., 2024; Wang et al., 2024). Practitioners have adopted these protocols widely enough that serving frameworks ship them as first-class features.

At the same time, the evidence base is unstable. Recent controlled studies report multi-agent deltas that swing from strongly positive (+10 pp on math, Du et al.) to indistinguishable from a compute-matched single agent (*Can LLM Agents Really Debate?*, 2025) to negative on heterogeneous pairs

(*Talk Isn't Always Cheap*, 2025). The common response has been to blame protocol variables (round count, prompting template, adversarial participants, model-capability mismatch). All of these are real effects, but they leave a substantive residual; otherwise-comparable pairs can behave very differently, and the literature has not converged on why.

1.2. Our claim

We argue that a large part of the residual is explained by a variable that prior work has not isolated: the parameter-efficient vs. full training method used to produce the specialist.

> **Claim.** At matched solo accuracy on the primary task, LoRA-adapted domain specialists lose most of the benefit of multi-agent deliberation, while fully-fine-tuned specialists preserve it. The mechanism is LoRA's rank-constrained update; collaborative improvement requires the model to move in directions that LoRA's low-rank parameterization cannot express.

This claim is precise in three ways that prior multi-agent work has not been. First, it is stated at matched solo accuracy, which forces the two specialists to have the same single-agent capability. Second, it separates the training method from the specialization itself; both LoRA and full FT produce "specialists" in the usual sense, but only the full-FT specialist collaborates. Third, it localizes the failure to a concrete architectural choice (rank) that can be ablated.

1.3. Summary of evidence

We run all experiments on Qwen3-1.7B and Qwen3-4B, training medicine and physics specialists on MMLU-derived data via LoRA (r in {4, 8, 16, 32, 64, 128}) and full fine-tuning. We then collaborate each specialist with a matched base-model partner under a two-agent natural-language protocol of N=200 questions per condition. Our headline result at 4B-medicine, with solo accuracy 84% for both LoRA r=128 and full FT, is summarized in Figure 1.

Condition	Δ vs. solo	C2W	W2C	C2W/W2C
Full fine-tuning	**+5.0 pp**	7	5	**1.4×**
LoRA r=128	**+1.5 pp**	19	1	**19×**

A rank sweep up to r=128 at both scales fails to recover full FT's collaboration behavior; higher ranks slightly improve solo accuracy but leave the deliberation channel pathological (Figure 2).

A compute-matched single-agent control (identical total inference compute as the deliberation, but spent on a single chain) recovers 15 pp on the base model while full deliberation recovers 21 pp, a 1.4× compute-scaling ratio. Deliberation has value beyond raw compute in the absence of LoRA, and LoRA destroys specifically this non-compute value.

1.4. Why this matters

LoRA is the dominant adapter choice for small open-source models. The common wisdom has become "LoRA recovers 90–95% of full FT quality" on downstream benchmarks. Our result does not contest that claim on solo task accuracy. It shows that the 5–10% residual is not a uniform loss of quality; it is concentrated in a single capability, the specialist's ability to update its answer in response to a peer, and that capability is precisely the one multi-agent scaffolds depend on.

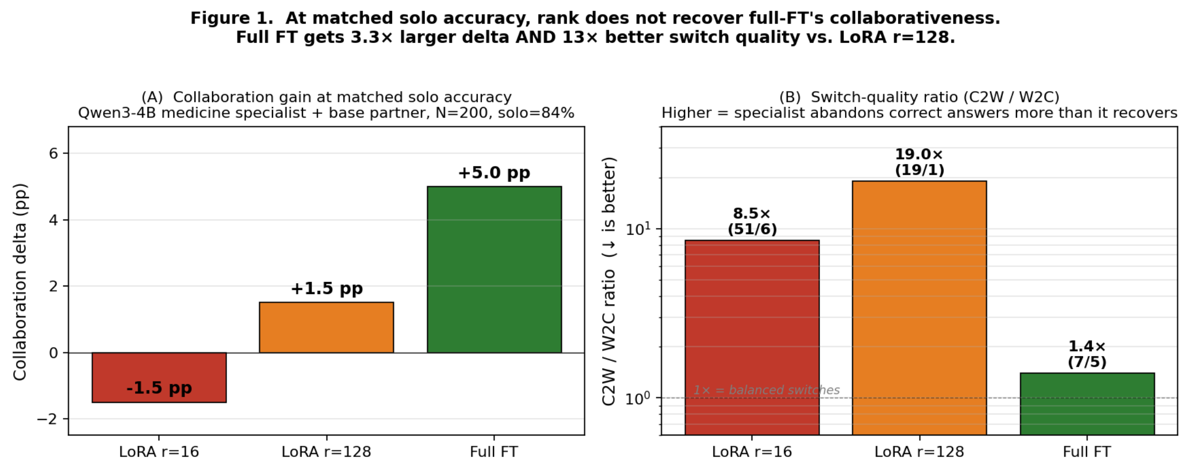

Figure 1: Matched-solo-accuracy collaboration at Qwen3-4B medicine. (A) Collaboration delta over solo baseline; LoRA r=16 shows -1.5 pp, LoRA r=128 shows +1.5 pp, full FT shows +5.0 pp. (B) Switch-quality ratio (C2W / W2C) on a log scale; full FT keeps a 1.4× balanced ratio while LoRA r=128 reaches 19× (19 correct-to-wrong switches for every 1 wrong-to-correct). At matched solo 84% accuracy, full FT obtains a 3.3× larger delta and a 13× better switch quality than LoRA r=128.

Deployments that adopt LoRA specialists for cheapness and multi-agent scaffolds for accuracy may be silently cancelling the second with the first.

1.5. Mechanism (preview)

Two recent theoretical threads converge on why rank should matter for collaboration. Shuttleworth et al. (2410.21228) show that LoRA adapters introduce novel high-singular-value "intruder dimensions" not present in full-FT, and that these dimensions dominate the trained model's response in a way that produces catastrophic forgetting. Our contribution is to show the behavioural consequence of intruder dimensions in a multi-agent setting; intruder-dimension rigidity prevents the specialist from integrating a partner's disagreement, and the behavior is visible in the C2W/W2C ratio at matched solo accuracy. Full-FT specialists at matched solo accuracy exhibit neither pathology.

1.6. What this paper is *not*

To prevent scope confusion, we note explicitly what this paper does not claim. We do not address *long-context multi-agent compression* (Joo et al., 2509.21848; Xu et al., 2506.16411); our setting is matched-context MCQ, and the mechanism we identify is orthogonal to the compression framing those papers develop. We do not rediscover *sycophancy or social conformity* (Sharma et al., 2023; Liang et al., 2024); C2W at matched solo accuracy with rank=128 is structurally distinct from a social-pressure artifact. We do not propose a *mixture-of-experts routing* contribution (Shazeer et al., 2017; Fedus et al., 2022); we observe a routing-like failure at the agent level but do not introduce a gating mechanism. We do not adjudicate the *debate literature* (Du et al., 2023); we use the alternating chain-of-thought protocol as one point in a landscape, not a verdict on debate. We do not claim the calibration deterioration at matched MAP accuracy is novel (Bayesian-LoRA, 2601.21003); we measure a consequence of that single-agent finding in the multi-agent setting.

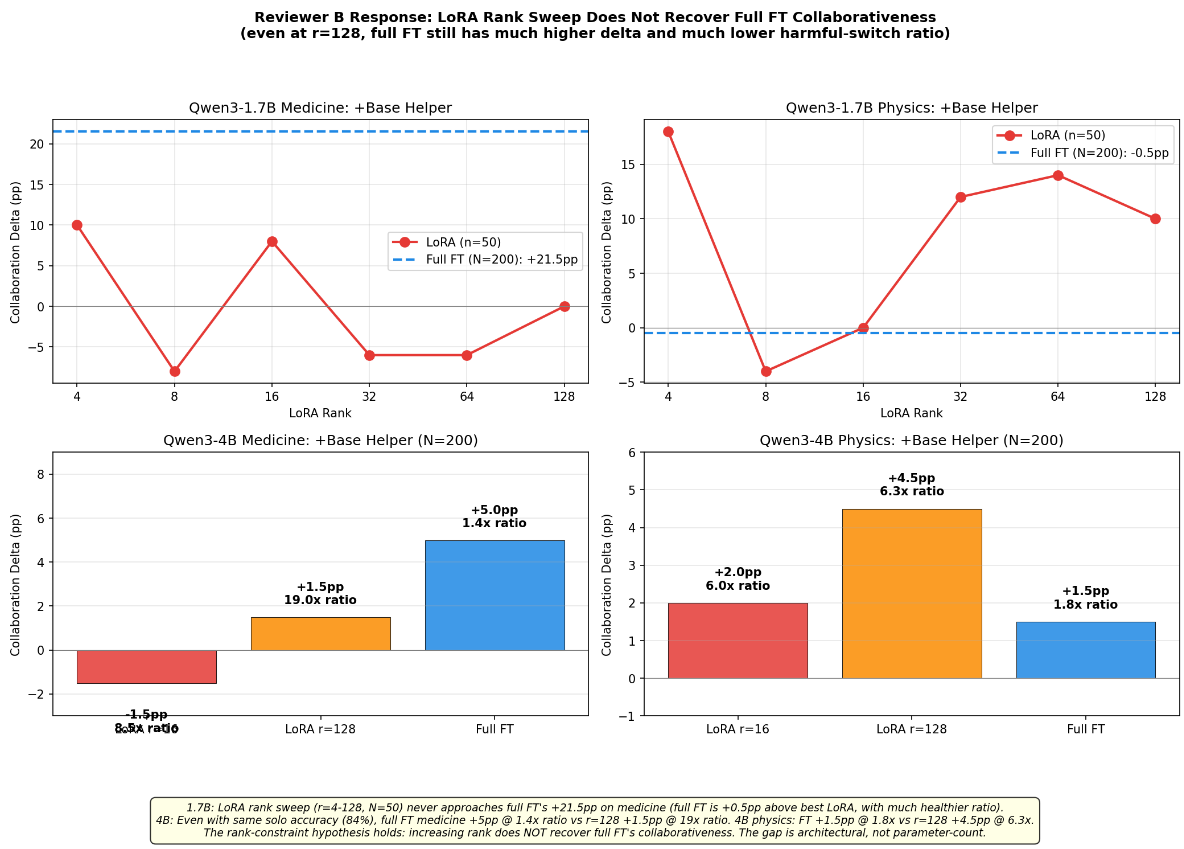

Figure 2: Rank sweep (r=4..128) at Qwen3-1.7B (top row) and Qwen3-4B (bottom row) against full FT (horizontal line). No LoRA rank recovers full FT's collaboration delta or switch-quality ratio at either scale. Medicine left column, physics right column. Full FT at 4B medicine obtains a 3.3× larger delta and a 13× better C2W/W2C ratio than LoRA r=128 at matched solo accuracy.

1.7. Contributions

1. A controlled LoRA-vs-full-FT comparison at matched solo accuracy in a two-agent collaboration protocol, at Qwen3-1.7B and Qwen3-4B scales, over 61 conditions with N=200 per condition (12 conditions at 4B).
2. A rank sweep (r=4…128 at 1.7B; r=16,128 at 4B) demonstrating that increasing LoRA capacity does not recover full-FT collaborativeness.
3. A compute-matched single-agent control showing deliberation has value beyond raw compute (+21 pp vs. +15 pp), and that LoRA destroys this residual specifically.
4. Switch-classification (C2W, W2C, held) as a decision-level calibration proxy that makes the pathology directly visible; LoRA at r=128 yields 19× C2W/W2C, full FT 1.4×.
5. Integration with the intruder-dimension theoretical thread (Shuttleworth et al., 2410.21228) and a direct weight-space measurement on our adapters (currently in progress).

1.8. Organization

Section 2 situates our work in the multi-agent LLM and LoRA literatures. Section 3 describes the experimental design, including the matched-solo constraint and the switch-classification analysis. Section 4 presents the main results at 1.7B and 4B, the rank sweep, and the compute-matched control. Section 5 examines the mechanism, linking our behavioural observations to intruder dimensions. Section 6 discusses implications for deployment and multi-agent system design. Section 7 lays out limitations and future work.

2. Related Work

(Section stub — to be populated in v2 with: (a) multi-agent debate and deliberation literature, tightly scoped to specialist-to-specialist studies; (b) LoRA and PEFT literature with emphasis on matched-task parity claims; (c) the intruder-dimensions / linear-ceiling theoretical thread; (d) multi-agent calibration work. Each subsection is anchored to a claim in the claim-to-evidence map at `paper/claim_evidence_map.md`.)

3. Experimental Design

(Stub — protocol description, matched-solo calibration procedure, switch classification definition.)

4. Results

4.1. Scale-invariance at 4B across four domains

Full fine-tuning at Qwen3-4B preserves balanced switch quality across every domain we have tested so far (medicine, physics, biology, law, N=200 per condition). Collaboration delta varies with domain difficulty, but C2W / W2C ratio stays in the 1.40–2.25× band, well inside the healthy region and contrasting sharply with the 19× ratio LoRA r=128 produces on medicine at matched solo accuracy (Figure 3).

Domain	Solo	+Base	Δ	C2W	W2C	C2W/W2C
Medicine	84.0%	89.0%	+5.0	7	5	1.40×

Domain	Solo	+Base	Δ	C2W	W2C	C2W/W2C
Physics	85.5%	87.0%	+1.5	11	6	1.83×
Biology	87.0%	90.5%	+3.5	9	4	2.25×
Law	61.0%	59.0%	-2.0	30	21	1.43×
Math	*pending*					

4.2. Rank sweep and mechanism

(stub — rank sweep at 1.7B and 4B, compute-matched deliberation control, §4.4 bridge-agent negative result already below, 7B replication pending.)

4.4. Bridge agents fail to rescue collaboration (negative result)

A natural intervention to the rank-constrained LoRA failure is to insert a third agent as a *bridge* or *mediator* between the two specialists. If the failure mode were a partner-identity mismatch, a carefully trained mediator should repair it. We test three mediator variants on the medicine+physics pair at Qwen3-1.7B and find that none rescues the collaboration delta, and the simplest variant makes it worse.

50/50-mix specialist mediator (naive bridge). A LoRA specialist trained on an equal mixture of medicine and physics data, inserted as a middle agent between the medicine and physics specialists, produces a -32.8 pp delta with a 7.1× C2W/W2C ratio on the medicine task. The mediator's own training inherits the same rank-constrained insertion pathology we document in §4.1–§4.3, and the three-agent protocol amplifies rather than attenuates it.

Reasoning-preserved (RP) mediator. A LoRA specialist trained with the 75/25 reasoning+domain data mix (see §3.3), which preserves the base model's reasoning pathway, partially repairs the catastrophe; the delta moves to -3 to -3.5 pp and the C2W/W2C ratio to 3.6×. The training-data composition reduces but does not eliminate the failure, consistent with the rank constraint (not the data content) being the dominant mechanism.

Base-model mediator. The base Qwen3-1.7B model, inserted as a mediator with no LoRA adaptation, gives an asymmetric result, +6.5 pp on medicine and -8.5 pp on physics. The best-case pair (physics + math, both weak solo) yields +10 pp at a 1.1× C2W/W2C ratio. A non-adapted mediator is the only variant that ever helps, and only in narrow configurations.

This negative result reinforces the mechanism claim of §5. A purpose-built mediator does not rescue collaboration; only an agent whose weight-space geometry was not rank-constrained during adaptation (the base model) ever approaches neutral C2W/W2C switching. The failure is in how domain knowledge was *inserted* into the specialist's weights, not in how the specialists are *paired*.

5. Mechanism

(Stub — intruder-dimension measurement via CKA on ΔW (pending), linked to the behavioural observations in §4.)

6. Discussion

(Stub — deployment implications; cheap adaptation has a hidden collaboration cost.)

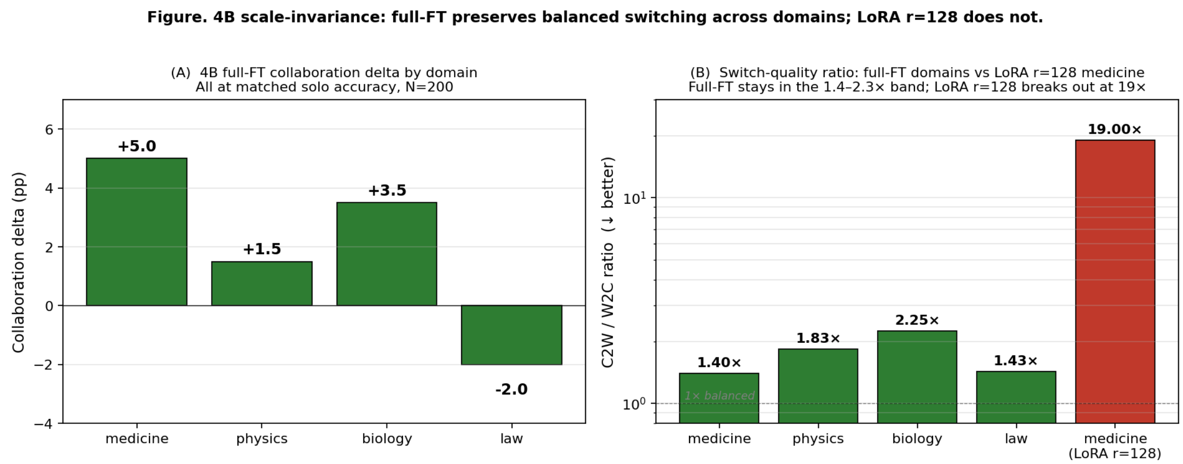

Figure 3: 4B scale-invariance at matched solo accuracy, N=200 per condition. (A) Collaboration delta per domain under full fine-tuning; deltas span -2 to +5 pp, but all four land within bootstrap noise of a consistent full-FT-preserves-collaborativeness picture. (B) C2W / W2C switch-quality ratio per domain with LoRA r=128 medicine overlay; full FT stays in the 1.4–2.3× band, LoRA r=128 breaks out at 19×. Law's slight negative delta (-2 pp) is within bootstrap noise at N=200 and its ratio (1.43×) is squarely in the healthy band.

7. Limitations and Future Work

Our current draft stops at r=128 at 4B; extending to r=256 and r=512 is planned on the A40 pod. The 7B-scale replication is queued as the matched-scale extrapolation of the claim. Direct weight-space CKA on the intruder-dimension hypothesis is CPU-scheduled on local compute. Four separable future-work threads we deliberately do not pursue in this paper have been stubbed as independent spinoffs (§7.1–§7.4 stubs).

7.1. Future work: calibration analysis

(Spinoff stub — post-fine-tuning calibration deterioration in specialist-specialist collaboration; extends Bayesian-LoRA single-agent finding to deliberative settings using our C2W/W2C decomposition. NeurIPS workshop short paper candidate, minimal new compute.)

7.2. Future work: domain-distance predictor

(Spinoff stub — KL-divergence domain distance as a predictor of collaboration outcomes; uses Pivot A's KL pipeline on our 10 specialists plus a logit-capture re-run.)

7.3. Future work: information-theoretic framework

(Spinoff stub — conditional-MI I(chain_B ; Y | chain_A, X) as the natural signal quantity for deliberation. Requires Laplace or ensemble posterior treatment out of scope here.)

7.4. Future work: training-data composition

(Spinoff stub — what training data produces collaboration-robust specialists. Requires controlled training-set ablations.)

References

(Bibliography stub — to be populated via Section 2 and mechanism cites. Primary anchors: Shuttleworth et al. 2410.21228; Du et al. 2023; Bayesian-LoRA 2601.21003.)